ZC-ASILD

PRODUCT REQUIREMENTS DOCUMENT (PRD)

ASIL-D Centralized Zone Controller

Automotive Zonal E/E Architecture | Rev A2 - Architecture, Atomic Requirements, Interfaces & Verification Baseline

Mission profile. The controller is a centralized zonal gateway that consolidates body, chassis and propulsion network routing for next-generation vehicle electrical/electronic architectures. The product baseline defines a multi-core lockstep MCU platform such as ST Stellar with PCIe routing fabric, AUTOSAR Adaptive with a POSIX-compliant operating environment, ASIL-D routing-integrity intent with strict Freedom From Interference, redundant 12 V / 48 V power inputs, an Ethernet backbone with hardware firewall and MACsec, an aluminum extruded chassis qualified to ISO 16750-3 vibration profiles, automated boundary scan, serialized MAC provisioning and OTA update with A/B rollback partitions.

Engineering baseline. This PRD defines the controlled zonal-controller architecture, atomic requirements, interface ownership, fault containment, safety mechanisms, cybersecurity controls and objective verification evidence needed for design review and release. Values that depend on vehicle topology, selected silicon, network loading, supplier data or installation are explicitly treated as controlled parameters until approved by the responsible engineering owner.

Definition	Controlled product interpretation
Product class	ASIL-D centralized zone controller / high-integrity zonal gateway for body, chassis and propulsion network routing in next-generation vehicle E/E architectures.
Safety / assurance	Source baseline: ASIL-D routing integrity with strict Freedom From Interference; ISO/SAE 21434 cybersecurity expansion; program-specific ISO 26262 allocation and evidence remain controlled by the vehicle safety lifecycle.
Lifecycle mission	15-year vehicle lifecycle with OTA A/B rollback, configuration genealogy, network-health/event logging, obsolescence control and evidence-based change assessment.
Primary architecture	Redundant 12 V / 48 V inputs -> protected power tree -> multi-core lockstep MCU / safety supervision -> PCIe routing fabric and high-speed Ethernet backbone -> CAN/LIN edge interfaces; AUTOSAR Adaptive / POSIX software; hardware firewall and MACsec; passive heatsink aluminum chassis.
Release philosophy	Hardware-independent routing, safety and security limits are separated from vehicle-variant, network-topology, supplier and calibration values. Evidence reuse is allowed only after documented similarity and impact review.

Release interpretation. Product requirements are separated from supplier-dependent, installation-dependent and calibration/configuration-controlled values. No planning target, typical silicon capability or unverified vehicle-network assumption shall be treated as a production limit without configuration-controlled evidence and approval.

PRODUCT GOVERNANCE, REQUIREMENT QUALITY & CHANGE CONTROL

Single-intent requirements, controlled parameters, explicit interface ownership and evidence-based release.

Safety and cybersecurity mechanisms are allocated by response time, independence and authority. Fast loss-of-routing-integrity, power, clock or reset faults are contained by hardware or safety-supervision paths where application software cannot be credited within the allocated FTTI. Supervisory software owns classification, logging, degraded routing, service reporting and controlled recovery. No network, diagnostic or OTA function may bypass local safety-state, power or resource-partitioning controls.

Requirements in this PRD are intentionally atomic. A requirement is considered closed only when the released hardware revision, software build, calibration / configuration set, network topology and test environment are identified by objective evidence. Compound descriptive paragraphs are supporting rationale, not substitutes for pass/fail requirements.

Parameter class	Examples / release rule
Immutable safety	Safe-state outputs, routing inhibit rules, watchdog / reset actions, memory-protection boundaries and independent fault containment. Change requires safety impact analysis and affected DVP regression.
Variant controlled	12 V / 48 V input option, connector set, network-port population, MCU / memory variant, software feature set and vehicle-domain allocation. Released by signed variant coding, BOM and configuration package.
Supplier dependent	MCU lockstep diagnostics, PCIe / Ethernet PHY limits, PMIC fault coverage, memory ECC behavior, transceiver EMC and thermal data. Use guaranteed production data and worst-case analysis; requalify on source change.
Installation dependent	Vehicle source impedance, grounding / bonding, harness length, Ethernet / CAN topology, cooling contact, mounting stiffness and EMC environment. Captured in vehicle ICD and validated at integration.
Calibration / policy	Routing tables, firewall rules, bandwidth / queue limits, watchdog thresholds, timeout maps, security policy and update permissions. Authenticated, range checked and traceable to verification evidence.

Traceability rule. Each product requirement shall trace upward to a product mission, hazard/risk control or external interface obligation and downward to at least one verification artifact. Evidence reuse after a processor, routing fabric, transceiver, OS, protocol, security policy or vehicle-network change is permitted only after documented similarity analysis proves the changed parameter remains inside the assumptions of the linked analyses; otherwise the impacted test, model and safety argument are reopened.

SYSTEM BOUNDARY, ASSUMPTIONS & EXTERNAL OWNERSHIP

The zone controller is one element of the vehicle E/E architecture. It receives redundant or multi-source vehicle power, terminates local CAN/LIN/Ethernet segments, forwards authorized traffic across the vehicle backbone and hosts safety- and non-safety software partitions. Central vehicle computer authority, external fuse / feeder protection, actuator-level safety mechanisms, remote ECU behavior and vehicle harness topology remain external unless explicitly allocated by the vehicle ICD.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZC-SYS-001	The released system boundary shall identify ownership of vehicle power protection, network termination, routing authority, wake/sleep control, time sources, diagnostics and external safety mechanisms.	System architecture and ICD review.
ZC-SYS-002	External mitigations credited in the safety case shall have explicit response-time, diagnostic and failure-independence assumptions.	Safety / ICD trace review.
ZC-SYS-003	Loss of an external permissive required for high-integrity routing shall transition affected traffic to the allocated non-permissive or degraded state without creating uncontrolled actuation authority.	HIL / network fault-injection test.
ZC-SYS-004	The source-defined centralized-zonal role shall not be interpreted as ownership of every vehicle function; functional ownership shall remain explicit in the released architecture.	Product and responsibility-allocation review.

Verification, margin and future-change rule

Changes to central-compute ownership, zone allocation, harness topology, actuator safety ownership or vehicle wake strategy reopen the interface safety and routing-authority analysis.

OPERATING MODES, STARTUP / SHUTDOWN & ROUTING AUTHORITY

The controller supports power-off, standby, wake, initialization, network-enabling, normal routing, degraded routing, diagnostic/service, update and controlled shutdown states. Route enable is permitted only after power, clocks, memory, safety supervision, security state and required network interfaces meet READY criteria. Reset, brownout or update recovery must not silently restore a previously unsafe or stale routing state.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZC-MOD-010	Each released operating mode shall define entry conditions, exit conditions, routing authority, interface availability, timeout behavior and safe / degraded fallback.	State-machine review and HIL transition matrix.
ZC-MOD-011	Transition from initialization or recovery to unrestricted routing shall require validated power, clock, memory, safety-monitor and configuration state.	Power-on / reset HIL test.
ZC-MOD-012	Reset, brownout or watchdog recovery shall not re-enable stale routing, firewall or actuator-command permissions without revalidation.	Brownout and reset fault injection.
ZC-MOD-013	Conflicting commands from central compute, service tools or local safety logic shall resolve to a deterministic non-permissive authority rule.	Command-conflict and privilege test.

Verification, margin and future-change rule

Any new state, wake source, diagnostic authority or routing-permission model requires state-machine, safety and cybersecurity regression.

REDUNDANT 12 V / 48 V INPUT, POWER TREE & PROTECTION

The product baseline includes redundant 12 V / 48 V power inputs. The released design defines which inputs are alternate variants versus concurrently redundant feeds, how cross-feed is prevented, how brownout and load transients are handled, and which power domains remain available in degraded operation. Power faults must not create uncontrolled network output or defeat both safety-monitoring paths through a shared rail or reset dependency.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZC-PWR-020	Each released power-input variant shall define normal, abnormal and transient voltage envelopes used for component, protection and reset design.	Electrical ICD and programmable-source sweep.
ZC-PWR-021	Power-path architecture shall prevent a single input fault from destructively back-feeding or disabling both independent safety / supervision paths beyond the allocated common-cause assumptions.	Power fault and cross-feed test.
ZC-PWR-022	Network transmit enable shall remain inhibited until required internal rails, clocks and reset supervisors meet READY criteria.	Power sequencing test.
ZC-PWR-023	Brownout and input-source switchover shall transition routing and software state without uncontrolled frame bursts, stale command replay or undefined outputs.	Brownout / source-transfer HIL test.

Verification, margin and future-change rule

Power-source, PMIC, feeder, connector or grounding changes reopen transient, cross-feed, reset and common-cause verification.

MULTI-CORE LOCKSTEP MCU, SAFETY ISLAND & COMPUTE PARTITIONING

The product baseline uses a multi-core lockstep MCU platform such as ST Stellar. Lockstep execution, safety island / monitor functions, memory protection, ECC, clock and reset diagnostics are used only to the extent supported by the selected device safety concept and program allocation. A processor diagnostic claim is not accepted without production data, configuration evidence and end-to-end fault-response timing.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZC-CMP-030	The released MCU configuration shall document lockstep mode, safety-monitor ownership, memory-protection regions, clock / reset diagnostics and startup self-test assumptions.	MCU safety-manual mapping and configuration review.
ZC-CMP-031	Detected core-lockstep or safety-monitor faults shall transition affected routing / software partitions to the allocated safe or degraded state within the program FTTI.	Processor fault-injection test.
ZC-CMP-032	A common clock, reset or power fault shall not invalidate all claimed independent monitoring without being detected and handled by the allocated architecture.	Common-cause analysis and fault injection.
ZC-CMP-033	MCU derivative or safety-manual revision changes shall require delta analysis against diagnostic coverage, timing and configuration assumptions.	Supplier-change safety review.

Verification, margin and future-change rule

MCU family, clock tree, safety-island configuration or core allocation changes reopen diagnostic coverage, timing and independence evidence.

PCIe ROUTING FABRIC, ENDPOINT ISOLATION & ERROR CONTAINMENT

The baseline architecture includes a PCIe routing fabric. PCIe is treated as an internal high-speed interconnect whose link training, address mapping, DMA authority, endpoint reset behavior and error containment must be configuration controlled. A failing or malicious endpoint must not gain unintended access to safety-relevant memory, routing tables or control registers.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZC-PCIe-040	The released PCIe topology shall define endpoint ownership, BAR / address windows, DMA permissions, reset domains and error-containment behavior.	PCIe architecture and configuration review.
ZC-PCIe-041	Uncorrectable link or endpoint errors shall be detected and contained so the fault cannot corrupt protected routing or safety data beyond the allocated boundary.	AER / link-fault injection test.
ZC-PCIe-042	DMA-capable endpoints shall be restricted by hardware protection so unauthorized memory or register access is rejected.	IOMMU / protection negative test as applicable.
ZC-PCIe-043	PCIe retraining or endpoint reset shall not restore stale permissions or cause uncontrolled vehicle-network output.	Link recovery HIL test.

Verification, margin and future-change rule

PCIe generation, switch, endpoint, DMA policy or memory-map changes require signal-integrity, performance, security and fault-containment regression.

AUTOMOTIVE ETHERNET BACKBONE & HIGH-INTEGRITY ROUTING

The product baseline uses an Ethernet backbone for centralized zonal routing. The product controls link state, ingress classification, forwarding, buffering and egress permissions so body, chassis and propulsion traffic remains deterministic within the released network budget. Routing integrity is defined by source/destination authority, data validity, bounded delay and safe behavior under overload or link failure, not by nominal line rate alone.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZC-ETH-050	Each Ethernet port shall have a released role, link mode, network / VLAN policy if used, bandwidth budget and failure fallback defined by the vehicle ICD.	Ethernet ICD and configuration review.
ZC-ETH-051	Routing overload shall be bounded so lower-criticality traffic cannot starve safety-relevant forwarding beyond the allocated latency / freshness budget.	Full-load and overload network test.
ZC-ETH-052	Loss, flap or error storm on one Ethernet link shall be contained to the allocated route set and shall not create uncontrolled message duplication.	Link-fault and storm injection.
ZC-ETH-053	Forwarding configuration shall be integrity protected and shall default to non-permissive behavior when configuration validity is lost.	Corrupt-configuration negative test.

Verification, margin and future-change rule

PHY, switch, topology, bandwidth allocation or routing-policy changes trigger latency, overload, EMC, safety and cybersecurity regression.

CAN / LIN EDGE INTERFACES & DETERMINISTIC GATEWAY OWNERSHIP

The detailed architecture expands the zonal-routing concept to explicit CAN and LIN edge-interface ownership. Receive, translate and forward behavior is defined per signal / frame, including units, freshness, timeout and failure semantics. Bus-off, stuck-dominant, malformed traffic or flood conditions cannot be allowed to consume unbounded compute or become a cross-domain actuation path.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZC-CAN-060	Each CAN / LIN interface shall define termination / transceiver ownership, supported bus mode, wake behavior and fail-silent output state.	Network ICD and bench conformance test.
ZC-CAN-061	Gateway rules shall define source, destination, transform, freshness / timeout and fallback for each safety- or performance-relevant routed data item.	Signal-routing trace review.
ZC-CAN-062	Bus-off, error flood or stuck-bus conditions shall be detected and contained without unbounded CPU / memory consumption or uncontrolled forwarding to other domains.	CAN / LIN fault-injection test.
ZC-CAN-063	Recovery from bus-off or transceiver reset shall not replay stale queued commands outside the released freshness window.	Bus recovery and stale-data test.

Verification, margin and future-change rule

Transceiver, bit rate, message set, translation rule or wake topology changes reopen signal-integrity, timing and safety evidence.

MEMORY, STORAGE, ECC & A/B PARTITION INTEGRITY

The zone controller depends on volatile memory, executable / calibration storage and the baseline A/B OTA rollback partitions. ECC, scrubbing, boot authentication, filesystem / storage integrity and power-fail behavior are treated as part of the routing-integrity argument. A valid rollback partition must be independently identifiable and protected from partial-update corruption.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZC-MEM-070	Safety- or routing-relevant RAM and nonvolatile storage shall use the selected MCU / memory protection mechanisms with error reporting and defined fault response.	Memory architecture and fault-injection review.
ZC-MEM-071	Detected uncorrectable memory corruption in protected data or executable regions shall prevent use of the affected content for unrestricted routing.	ECC / corruption fault injection.
ZC-MEM-072	A/B software partitions shall have independently verifiable image identity, integrity state and rollback metadata protected against power interruption.	Power-fail update and rollback test.
ZC-MEM-073	Storage endurance and error rates shall be included in the released lifecycle model for logging, update and configuration usage.	Endurance analysis and accelerated test.

Verification, margin and future-change rule

Memory technology, density, ECC scheme, filesystem or partition layout changes reopen integrity, update and lifecycle evidence.

AUTOSAR ADAPTIVE / POSIX SOFTWARE ARCHITECTURE

The product baseline specifies AUTOSAR Adaptive integrated with a POSIX-compliant OS for service-oriented routing. Software architecture decomposes lifecycle management, routing services, diagnostics, security, network management and platform services with explicit startup order, resource budgets and failure containment. Dynamic features are permitted only within bounded, monitored and recoverable resource limits appropriate to the safety allocation.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZC-SW-080	The released software architecture shall identify service ownership, startup / shutdown dependencies, execution context, inter-process communication and safety relevance.	Software architecture and traceability review.
ZC-SW-081	Maximum CPU, memory, interrupt and network-service load shall retain a release-controlled execution reserve under defined fault and diagnostic activity.	WCET / load stress measurement.
ZC-SW-082	Failure or restart of a non-safety service shall not corrupt safety-relevant routing configuration or violate the allocated FFI boundary.	Process crash / restart fault injection.
ZC-SW-083	OS, compiler, middleware or AUTOSAR platform changes shall require timing, compatibility, safety and cybersecurity impact analysis.	Software platform change review.

Verification, margin and future-change rule

OS, AUTOSAR release, compiler, scheduler or middleware changes reopen WCET, resource, FFI and cybersecurity evidence.

FREEDOM FROM INTERFERENCE & MIXED-CRITICALITY RESOURCE CONTROL

The product baseline explicitly requires strict Freedom From Interference for ASIL-D routing integrity. FFI covers memory, CPU time, interrupts, communication buffers, DMA, shared drivers, clocks, power / reset and configuration authority. Safety claims shall not rely on conceptual software separation if a lower-criticality workload can exhaust or corrupt a shared physical resource.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZC-FFI-090	The FFI concept shall identify every shared resource between safety-relevant and lower-criticality partitions and the enforced protection mechanism.	FFI architecture and safety review.
ZC-FFI-091	Lower-criticality CPU, memory, interrupt or network load shall be bounded so it cannot violate the released safety-partition timing or data-integrity budget.	Resource-exhaustion stress test.
ZC-FFI-092	Unauthorized cross-partition memory, DMA, device-register or routing-table access shall be blocked by hardware / privileged software controls.	Protection negative test.
ZC-FFI-093	A failure in a shared driver or platform service shall transition affected safety functions to a defined degraded or safe state rather than silently bypass FFI.	Shared-resource fault injection.

Verification, margin and future-change rule

Partition map, shared driver, DMA client, interrupt architecture or resource-budget change reopens the FFI argument and stress DVP.

FUNCTIONAL SAFETY, ASIL-D ROUTING INTEGRITY & FTTI

The product baseline assigns ASIL-D routing-integrity intent. The detailed safety lifecycle must derive hazards, safety goals, functional / technical safety requirements, safe / degraded states, FTTI and diagnostic coverage from the actual vehicle architecture. The controller shall not claim ASIL-D at box level without the allocated system assumptions, independence arguments and verification evidence.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZC-SAF-100	Each safety objective allocated to the zone controller shall trace from vehicle hazard / safety analysis to product requirement, mechanism and verification evidence.	Safety traceability review.
ZC-SAF-101	FTTI and fault-response budgets shall include detection, communication, software scheduling, reset / isolation and downstream actuator response where credited.	End-to-end timing analysis and fault test.
ZC-SAF-102	Single-point, latent and common-cause faults shall be analyzed for compute, power, memory, network and supervision paths used by safety-relevant routing.	FMEDA / DFA / safety analysis review.
ZC-SAF-103	The controller shall not claim vehicle-level risk reduction from an external ECU or actuator mechanism unless that dependency is explicit in the ICD and verified.	System safety consistency audit.

Verification, margin and future-change rule

Vehicle hazard allocation, route criticality, redundancy or downstream-actuator assumptions changing requires safety reallocation before evidence reuse.

WATCHDOG, CLOCK / RESET SUPERVISION & LATENT-FAULT DETECTION

Routing integrity depends on independent monitoring of application execution, clocks, reset sequencing, power-good status and latent safety mechanisms. Watchdog servicing must be tied to meaningful execution milestones, not merely a periodic task. Reset causes and first-fault information are preserved so repeated faults do not become an uncontrolled reboot loop.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZC-SUP-110	The released watchdog architecture shall define independent monitor, service milestones, timeout budget and resulting safe / degraded action.	Watchdog architecture and timing review.
ZC-SUP-111	Clock, reset and power-supervisor faults that can invalidate routing integrity shall be detected or forced to the allocated safe state within the program FTTI.	Clock / reset / power fault injection.
ZC-SUP-112	Watchdog service shall fail if required routing / safety execution milestones are not completed, even when a low-level tick continues.	Processor-stall and task-starvation test.
ZC-SUP-113	Reset-loop protection shall limit automatic restart and preserve cause / first-fault data for service diagnosis.	Repeated reset / fault test.

Verification, margin and future-change rule

Watchdog IC, clock source, reset tree, safety monitor or startup-sequence changes trigger supervision and timing regression.

CYBERSECURITY, HARDWARE FIREWALL & MACsec

The product baseline specifies an Ethernet backbone with hardware firewall and MACsec. Cybersecurity is integrated with the safety authority model: secure boot, identities, firewall policy, key lifecycle, link protection, maintenance access and network segmentation cannot delay local safety reactions. Loss of trust causes non-permissive behavior for protected traffic, not continued routing on stale authorization.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZC-CYB-120	Trust anchors, secure-boot policy, hardware-firewall rules, MACsec policy and maintenance roles shall be configuration controlled and traceable to the released variant.	Cybersecurity architecture and configuration review.
ZC-CYB-121	Only authenticated / authorized software and network-policy configuration shall enable protected routing in normal vehicle operation.	Modified-image / policy negative test.
ZC-CYB-122	Loss or invalidity of security services shall reject untrusted traffic / configuration without inhibiting autonomous local safety monitoring and safe-state action.	Concurrent cyber-fault / safety test.
ZC-CYB-123	Keys, certificates and security-relevant identifiers shall have a controlled provisioning, rotation, revocation and service lifecycle.	Key-management and provisioning audit.

Verification, margin and future-change rule

Firewall silicon, MACsec implementation, trust anchor, key hierarchy, service-role or network-segmentation changes reopen threat and safety interaction analysis.

OTA UPDATE, A/B ROLLBACK & CONFIGURATION AUTHORITY

The 15-year lifecycle baseline includes OTA capability with A/B rollback partitions. Update authority covers application software, platform software, routing / firewall policy and calibration only through authenticated, variant-compatible and power-fail-safe mechanisms. A failed update must not leave the controller in an ambiguous image / policy state or restore configuration incompatible with the installed hardware.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZC-OTA-130	Each OTA package shall be authenticated, variant compatible and traceable to a released software / configuration manifest before activation.	Update package and manifest negative test.
ZC-OTA-131	A/B update shall preserve at least one bootable, authenticated fallback image and consistent rollback metadata across power interruption.	Power-fail update / rollback test.
ZC-OTA-132	Rollback shall restore a compatible routing, firewall and calibration set rather than software image alone.	Configuration-consistency rollback test.
ZC-OTA-133	Update activation shall occur only in released vehicle / power states and shall not create uncontrolled network transmission during transition.	Vehicle-state and power interruption test.

Verification, margin and future-change rule

Bootloader, storage layout, update transport, signing scheme or rollback policy changes trigger security, power-fail and compatibility regression.

POWER DISSIPATION, PASSIVE HEATSINK & HOT-SOAK DERATING

The product baseline uses passive heatsink cooling with redundant 12 V / 48 V input capability. Thermal design covers MCU / SoC, PMIC, memories, Ethernet / PCIe devices and transceivers across network load, ambient, installation interface and end-of-life thermal resistance. Thermal throttling may reduce noncritical service capacity but cannot silently violate the timing budget of safety-relevant routing.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZC-THM-140	The released thermal model shall define component loss, ambient / mounting boundary, interface resistance and end-of-life thermal margin for each variant.	Thermal model and worst-case analysis.
ZC-THM-141	Hotspot / junction estimates shall remain below released limits under maximum defined traffic and compute load including degraded thermal interface.	Thermal mapping and traffic stress test.
ZC-THM-142	Thermal derating shall preserve safety-relevant routing timing or transition explicitly to the allocated degraded state before timing margin is exhausted.	Hot-soak derating HIL test.
ZC-THM-143	Invalid temperature data shall cause conservative power / performance handling and diagnostic reporting rather than unrestricted operation.	Sensor fault-injection test.

Verification, margin and future-change rule

Heatsink, TIM, enclosure, processor power, mounting or thermal-interface changes reopen hot-soak and timing-margin verification.

MECHANICAL PACKAGING, ALUMINUM CHASSIS & VEHICLE INTERFACES

The product baseline specifies an aluminum extruded chassis tested to ISO 16750-3 vibration profiles. The released package defines mounting, connector retention, PCB support, grounding / bonding, thermal interfaces and tolerance stack-up so vibration or assembly variation does not degrade high-speed links, power integrity or safety monitoring. Any ingress or environmental sealing requirement remains program-specific unless separately allocated.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZC-MECH-150	The enclosure, PCB support, mounting and connector system shall be configuration controlled to the released vibration and thermal environment.	Mechanical drawing / stack-up review.
ZC-MECH-151	Post-vibration electrical, network and safety-monitor performance shall remain within released limits without intermittent connector or solder-joint behavior.	ISO 16750-3 aligned sequential vibration / functional test.
ZC-MECH-152	Grounding / bonding and shield termination shall support the released EMC and high-speed signal-integrity assumptions.	Mechanical / EMC interface review.
ZC-MECH-153	Mechanical tolerance and thermal expansion shall not consume connector engagement, heatsink contact or high-speed channel margin below the released minimum.	Tolerance analysis and environmental correlation.

Verification, margin and future-change rule

Chassis material, mounting, PCB stack-up, connector or shield-termination changes reopen vibration, thermal and SI/EMC evidence.

MANUFACTURING, BOUNDARY SCAN & SERIALIZED MAC PROVISIONING

The product baseline requires automated boundary scan and serialized MAC provisioning. Production controls critical-to-quality characteristics across power rails, reset, high-speed interfaces, safety I/O, memory, network ports and identity programming. EOL configuration must prevent incompatible hardware, firmware, MAC address, security identity or vehicle variant combinations from being released.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZC-MFG-160	EOL test shall verify safety-relevant power, reset, memory, network-interface and supervision functions with objective pass/fail evidence.	EOL fixture qualification and golden-unit correlation.
ZC-MFG-161	Automated boundary scan / structural test shall cover the released accessible interconnect set and identify coverage gaps requiring alternate test methods.	Boundary-scan coverage audit.
ZC-MFG-162	MAC addresses and security / manufacturing identities shall be unique, serialized and protected against duplicate or unauthorized provisioning.	Provisioning process audit.
ZC-MFG-163	Software trim or routing calibration shall not be used to mask an assembly defect that consumes safety, SI or thermal margin.	Process capability and calibration review.

Verification, margin and future-change rule

PCB revision, test-access change, provisioning backend, serializer or identity schema changes require manufacturing / EOL requalification.

NETWORK MANAGEMENT, DIAGNOSTICS & DEGRADED ROUTING

A centralized gateway must distinguish physical-link faults, protocol faults, configuration faults, safety-supervision faults and cybersecurity denials while preserving first-fault context. Diagnostics support service decisions without gaining unrestricted routing authority. Degraded routing is explicit: allowed routes, traffic classes, duration and recovery criteria are configuration controlled rather than improvised after a fault.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZC-DIAG-170	Fault classification shall distinguish power, compute, memory, link, protocol, routing, safety and cybersecurity causes with traceable status.	Diagnostic coverage and fault-injection review.
ZC-DIAG-171	First-fault data shall preserve relevant power state, reset cause, network state, software / configuration identity and timing context across controlled reboot.	Power-fail and event-log test.
ZC-DIAG-172	Degraded routing shall permit only the released route / traffic subset and shall not increase command authority relative to normal operation.	Degraded-mode network test.
ZC-DIAG-173	Diagnostic / service requests shall be rate limited and privilege controlled so they cannot starve safety-relevant processing or bypass routing policy.	Diagnostic load and authorization test.

Verification, margin and future-change rule

Diagnostic protocol, degraded-route set, event-log schema or service-role changes reopen timing, cybersecurity and safety regression.

15-YEAR LIFECYCLE, OTA SUPPORT & OBSOLESCENCE CONTROL

The lifecycle baseline targets long-term OTA support with A/B rollback. The released lifecycle model extends that intent to processor / memory wear, connector and solder fatigue, thermal cycling, storage endurance, update frequency, security-certificate life, supplier obsolescence and field-data governance. A long service claim requires a mission profile and evidence, not only nominal component ratings.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZC-LIFE-180	The released mission profile shall define vehicle operating time, temperature / vibration exposure, network activity, update count and service events used for lifecycle analysis.	Mission-profile and reliability-plan review.
ZC-LIFE-181	NVM / storage usage for logging, A/B images and configuration shall remain within released endurance with corruption recovery over the lifecycle.	Endurance and power-interruption testing.
ZC-LIFE-182	Field health / event data used for maintenance or reliability decisions shall be configuration controlled, privacy / security governed and traceable to software / hardware identity.	Field-data governance audit.
ZC-LIFE-183	Supplier or component obsolescence substitutions shall require similarity analysis against power, timing, SI/EMC, thermal, safety, cybersecurity and lifecycle assumptions.	Change-control review.

Verification, margin and future-change rule

Any mission-profile expansion or supplier substitution beyond qualified similarity bounds triggers targeted reliability, safety, cyber and DVP regression.

PART II - DETAILED SUBSYSTEM PRDS & ENGINEERING ASSURANCE ANNEX

Expanded architecture, atomic requirements, fault containment, interfaces, DVP and lifecycle evidence.

This annex extends the product-level zone-controller baseline into subsystem requirements for design review, hardware/software allocation, ISO 26262 work products, cybersecurity analysis, interface definition, production release and lifecycle change control. Controlled parameters are used where the final vehicle zone allocation, 12 V / 48 V power architecture, harness topology, network population, processor / memory variant, cooling interface, software load or supplier configuration has not fixed a value.

Across Part II, every TBC shall have an owner and closure method. Evidence shall identify the released configuration, calibrated instrumentation and raw records needed to demonstrate the mechanism. Prior evidence may be reused only after documented similarity and impact review shows the affected stress, timing, independence and margin remain bounded; otherwise the impacted analysis and DVP shall be repeated.

Annex family	Primary closure intent
A01-A05	Power-on state, dual input power, transients, rail sequencing, lockstep compute
A06-A10	Safety supervision, memory integrity, PCIe containment, Ethernet PHY, routing load
A11-A15	CAN/LIN gateway, signal validity, AUTOSAR lifecycle, POSIX timing, FFI
A16-A20	Time / freshness, secure boot, MACsec, firewall, OTA rollback
A21-A25	Diagnostics, thermal, EMC, harness / SI, mechanical vibration
A26-A30	Manufacturing, provisioning, full-load DVP, environmental stress, lifecycle / second source

VEHICLE POWER-ON & NETWORK-ENABLE STATE MACHINE

Detailed subsystem PRD / engineering assurance annex - control closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA01-01	The Vehicle Power-On and Network-Enable State Machine function shall configuration-control state authority, command and saturation limits, valid input conditions, timing bounds and recovery behavior for every released variant.	Architecture / ICD / configuration review.
ZCA01-02	Controller states, route-enable conditions, reset / wake timing and invalid-input behavior shall be bounded across the released power, network and software-startup envelope.	SIL / HIL state-transition sweep with brownout, reset and invalid-configuration injection.
ZCA01-03	Invalid state, timing, resource or input information for Vehicle Power-On and Network-Enable State Machine shall restrict the affected service or route and shall not permit stale configuration or uncontrolled transmit authority.	SIL/HIL timing, saturation, overload and invalid-input boundary sweep.

Verification, margin and future-change rule

Incorrect sequencing can expose network transmit authority before power, memory, safety or security readiness is qualified. Closure evidence shall include: SIL / HIL state-transition sweep with brownout, reset and invalid-configuration injection. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

DUAL 12 V / 48 V INPUT OR-ING & CROSS-FEED CONTAINMENT

Detailed subsystem PRD / engineering assurance annex - power closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA02-01	The Dual 12 V/48 V Input OR-ing and Cross-Feed Containment design shall configuration-control the operating envelope, interfaces, ownership, limits, assumptions, fault response and objective release evidence for every released variant.	Architecture / ICD / configuration review.
ZCA02-02	A single input-source fault or source-transfer event shall not destructively back-feed the alternate supply or invalidate both safety-monitoring paths beyond the allocated common-cause assumptions.	Programmable dual-source fault matrix with cross-feed current, reset and routing-state capture.
ZCA02-03	If qualified margin or required health data for Dual 12 V/48 V Input OR-ing and Cross-Feed Containment is unavailable, the controller shall inhibit or restrict the affected network/compute function and enter the allocated non-permissive or degraded behavior.	Boundary-condition and fault-injection / release-gate test with physical or network response recorded.

Verification, margin and future-change rule

A redundant-input label is unsafe unless transfer, common-cause and fault isolation behavior are demonstrated end to end. Closure evidence shall include: Programmable dual-source fault matrix with cross-feed current, reset and routing-state capture. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

CRANK / LOAD-DUMP / REVERSE-POLARITY & BROWNOUT

Detailed subsystem PRD / engineering assurance annex - power closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA03-01	The Crank/Load-Dump/Reverse-Polarity and Brownout design shall configuration-control the operating envelope, interfaces, ownership, limits, assumptions, fault response and objective release evidence for every released variant.	Architecture / ICD / configuration review.
ZCA03-02	Normal, abnormal and transient supply conditions shall map to deterministic continue / degrade / reset behavior without uncontrolled transmit bursts or stale route restoration.	Automotive supply transient sweep with network traffic and first-fault logging.
ZCA03-03	If qualified margin or required health data for Crank/Load-Dump/Reverse-Polarity and Brownout is unavailable, the controller shall inhibit or restrict the affected network/compute function and enter the allocated non-permissive or degraded behavior.	Boundary-condition and fault-injection / release-gate test with physical or network response recorded.

Verification, margin and future-change rule

Vehicle supply transients can create intermittent logic behavior that is more dangerous than a clean power loss. Closure evidence shall include: Automotive supply transient sweep with network traffic and first-fault logging. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

PMIC RAIL SEQUENCING, POWER-GOOD & RESET TREE

Detailed subsystem PRD / engineering assurance annex - hardware closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA04-01	The PMIC Rail Sequencing, Power-Good and Reset Tree design shall configuration-control its electrical operating envelope, component and interface limits, protection thresholds, tolerance assumptions and end-of-life margin for every released hardware variant.	Architecture / ICD / configuration review.
ZCA04-02	Internal rail sequencing and reset release shall meet device / memory / transceiver dependencies and shall hold network transmit paths non-permissive until required domains are valid.	Oscilloscope sequencing test plus rail UV / OV and reset fault injection.
ZCA04-03	If qualified margin or required health data for PMIC Rail Sequencing, Power-Good and Reset Tree is unavailable, the controller shall inhibit or restrict the affected network/compute function and enter the allocated non-permissive or degraded behavior.	Representative hardware fault-injection test with network and safety response recorded.

Verification, margin and future-change rule

Shared PMIC or reset dependencies can defeat claimed independence if an invalid partial-power state is not contained. Closure evidence shall include: Oscilloscope sequencing test plus rail UV / OV and reset fault injection. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

LOCKSTEP MCU CORE SUPERVISION & DEGRADED COMPUTE

Detailed subsystem PRD / engineering assurance annex - safety closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA05-01	The Lockstep MCU Core Supervision and Degraded Compute safety mechanism shall configuration-control hazard-relevant limits, independent detection and shutdown paths, FTTI assumptions, diagnostic coverage, safe/degraded states and external dependencies.	Architecture / ICD / configuration review.
ZCA05-02	Detected lockstep / core faults shall transition the affected compute partition to the allocated safe / degraded route set within the program FTTI and without uncontrolled message replay.	MCU fault injection with network latency / output monitoring.
ZCA05-03	Loss of required margin, diagnostic validity or protection readiness for Lockstep MCU Core Supervision and Degraded Compute shall force the allocated safe/degraded routing state within the released FTTI without dependence on a lower-integrity workload.	Safety-mechanism fault injection with FTTI and independence evidence.

Verification, margin and future-change rule

Processor diagnostics are useful only when their detection path and physical routing response are bounded in time. Closure evidence shall include: MCU fault injection with network latency / output monitoring. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

SAFETY ISLAND, EXTERNAL WATCHDOG & CLOCK MONITORING

Detailed subsystem PRD / engineering assurance annex - safety closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA06-01	The Safety Island, External Watchdog and Clock Monitoring safety mechanism shall configuration-control hazard-relevant limits, independent detection and shutdown paths, FTTI assumptions, diagnostic coverage, safe/degraded states and external dependencies.	Architecture / ICD / configuration review.
ZCA06-02	The independent supervision path shall detect or force safe behavior for application stall, invalid clock and reset faults within the released timing allocation.	Processor stall, clock fault and watchdog-service fault injection.
ZCA06-03	Loss of required margin, diagnostic validity or protection readiness for Safety Island, External Watchdog and Clock Monitoring shall force the allocated safe/degraded routing state within the released FTTI without dependence on a lower-integrity workload.	Safety-mechanism fault injection with FTTI and independence evidence.

Verification, margin and future-change rule

A watchdog that can be serviced by a failing task does not provide credible independence. Closure evidence shall include: Processor stall, clock fault and watchdog-service fault injection. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

DDR / SRAM ECC, FLASH INTEGRITY & MEMORY SCRUBBING

Detailed subsystem PRD / engineering assurance annex - hardware closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA07-01	The DDR/SRAM ECC, Flash Integrity and Memory Scrubbing design shall configuration-control its electrical operating envelope, component and interface limits, protection thresholds, tolerance assumptions and end-of-life margin for every released hardware variant.	Architecture / ICD / configuration review.
ZCA07-02	Uncorrectable corruption of safety- or routing-relevant memory shall be detected before affected data is used for unrestricted forwarding.	ECC injection, memory corruption and boot-integrity test.
ZCA07-03	If qualified margin or required health data for DDR/SRAM ECC, Flash Integrity and Memory Scrubbing is unavailable, the controller shall inhibit or restrict the affected network/compute function and enter the allocated non-permissive or degraded behavior.	Representative hardware fault-injection test with network and safety response recorded.

Verification, margin and future-change rule

Latent memory corruption can preserve apparently valid execution while silently changing routing or policy data. Closure evidence shall include: ECC injection, memory corruption and boot-integrity test. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

PCIe LINK TRAINING, AER & ENDPOINT FAULT CONTAINMENT

Detailed subsystem PRD / engineering assurance annex - interface closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA08-01	The PCIe Link Training, AER and Endpoint Fault Containment interface shall configuration-control signal ownership, units/range, update rate, freshness/integrity checks, timeout behavior, authority and non-permissive fallback.	Architecture / ICD / configuration review.
ZCA08-02	PCIe endpoint / link faults shall be contained so an affected endpoint cannot corrupt protected memory or produce uncontrolled vehicle-network traffic during recovery.	Link-error / endpoint-reset / DMA negative test with traffic capture.
ZCA08-03	Invalid, stale or conflicting information for PCIe Link Training, AER and Endpoint Fault Containment shall be treated as non-permissive; the controller shall restrict the affected route or service without granting additional network authority.	Protocol conformance with timeout, stale-data, range, integrity and authority fault injection.

Verification, margin and future-change rule

A high-speed internal interconnect can turn a local endpoint fault into a system fault unless address, DMA and reset boundaries are enforced. Closure evidence shall include: Link-error / endpoint-reset / DMA negative test with traffic capture. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

ETHERNET PHY, LINK LOSS & PHYSICAL-LAYER EMC

Detailed subsystem PRD / engineering assurance annex - interface closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA09-01	The Ethernet PHY, Link Loss and Physical-Layer EMC interface shall configuration-control signal ownership, units/range, update rate, freshness/integrity checks, timeout behavior, authority and non-permissive fallback.	Architecture / ICD / configuration review.
ZCA09-02	Ethernet link loss, flap or physical-layer error shall transition affected routes according to the released timeout / degraded policy without uncontrolled duplication.	Link-interruption and EMC susceptibility test with route-state logging.
ZCA09-03	Invalid, stale or conflicting information for Ethernet PHY, Link Loss and Physical-Layer EMC shall be treated as non-permissive; the controller shall restrict the affected route or service without granting additional network authority.	Protocol conformance with timeout, stale-data, range, integrity and authority fault injection.

Verification, margin and future-change rule

Physical-layer intermittency can create stale or duplicated traffic before higher layers declare a clean link failure. Closure evidence shall include: Link-interruption and EMC susceptibility test with route-state logging. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

ROUTING TABLE INTEGRITY, BANDWIDTH & QUEUE CONTAINMENT

Detailed subsystem PRD / engineering assurance annex - control closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA10-01	The Routing Table Integrity, Bandwidth and Queue Containment function shall configuration-control state authority, command and saturation limits, valid input conditions, timing bounds and recovery behavior for every released variant.	Architecture / ICD / configuration review.
ZCA10-02	The released forwarding policy shall bound queue occupancy and latency so lower-criticality load cannot violate safety-relevant freshness / timing requirements.	Traffic-generator full-load and overload sweep with queue / latency instrumentation.
ZCA10-03	Invalid state, timing, resource or input information for Routing Table Integrity, Bandwidth and Queue Containment shall restrict the affected service or route and shall not permit stale configuration or uncontrolled transmit authority.	SIL/HIL timing, saturation, overload and invalid-input boundary sweep.

Verification, margin and future-change rule

A gateway can meet nominal throughput yet fail safety timing when buffers, queues or shared egress ports saturate. Closure evidence shall include: Traffic-generator full-load and overload sweep with queue / latency instrumentation. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

CAN / LIN EDGE GATEWAY, BUS-OFF & FLOOD CONTROL

Detailed subsystem PRD / engineering assurance annex - interface closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA11-01	The CAN/LIN Edge Gateway, Bus-Off and Flood Control interface shall configuration-control signal ownership, units/range, update rate, freshness/integrity checks, timeout behavior, authority and non-permissive fallback.	Architecture / ICD / configuration review.
ZCA11-02	Bus-off, stuck-bus or frame-flood conditions shall be detected and contained without unbounded CPU / memory load or unsafe forwarding into other domains.	CAN / LIN bus-off, dominant, flood and recovery fault injection.
ZCA11-03	Invalid, stale or conflicting information for CAN/LIN Edge Gateway, Bus-Off and Flood Control shall be treated as non-permissive; the controller shall restrict the affected route or service without granting additional network authority.	Protocol conformance with timeout, stale-data, range, integrity and authority fault injection.

Verification, margin and future-change rule

A fault on a low-speed edge bus can become a centralized resource-exhaustion failure if gateway intake is not bounded. Closure evidence shall include: CAN / LIN bus-off, dominant, flood and recovery fault injection. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

SIGNAL OWNERSHIP, FRESHNESS, RANGE & TIMEOUT

Detailed subsystem PRD / engineering assurance annex - interface closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA12-01	The Signal Ownership, Freshness, Range and Timeout interface shall configuration-control signal ownership, units/range, update rate, freshness/integrity checks, timeout behavior, authority and non-permissive fallback.	Architecture / ICD / configuration review.
ZCA12-02	Every safety- or performance-relevant routed data item shall define source, destination, encoding / range, freshness / integrity rule, timeout and fallback behavior.	Protocol conformance with stale, duplicate, range and authority fault injection.
ZCA12-03	Invalid, stale or conflicting information for Signal Ownership, Freshness, Range and Timeout shall be treated as non-permissive; the controller shall restrict the affected route or service without granting additional network authority.	Protocol conformance with timeout, stale-data, range, integrity and authority fault injection.

Verification, margin and future-change rule

Stale or ambiguously owned data can be more dangerous than complete communication loss because it may remain plausible. Closure evidence shall include: Protocol conformance with stale, duplicate, range and authority fault injection. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

AUTOSAR ADAPTIVE LIFECYCLE & SERVICE STARTUP

Detailed subsystem PRD / engineering assurance annex - software closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA13-01	The AUTOSAR Adaptive Lifecycle and Service Startup software function shall configuration-control task ownership, scheduling and WCET bounds, data validity, state transitions, diagnostics, watchdog interaction and recovery behavior.	Architecture / ICD / configuration review.
ZCA13-02	Adaptive applications and platform services shall expose routing functionality only after dependencies and health state meet the released lifecycle criteria.	Service-start / crash / restart HIL with route availability monitoring.
ZCA13-03	Invalid state, timing, resource or input information for AUTOSAR Adaptive Lifecycle and Service Startup shall restrict the affected service or route and shall not permit stale configuration or uncontrolled transmit authority.	SIL/HIL timing, resource-exhaustion and invalid-state injection.

Verification, margin and future-change rule

Service-oriented startup can create race conditions if route availability is not tied to dependency and health readiness. Closure evidence shall include: Service-start / crash / restart HIL with route availability monitoring. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

POSIX SCHEDULING, WCET, CPU & MEMORY RESERVE

Detailed subsystem PRD / engineering assurance annex - control closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA14-01	The POSIX Scheduling, WCET, CPU and Memory Reserve function shall configuration-control state authority, command and saturation limits, valid input conditions, timing bounds and recovery behavior for every released variant.	Architecture / ICD / configuration review.
ZCA14-02	Measured WCET, CPU / interrupt load and memory use shall retain release-controlled reserve at maximum defined network, diagnostic and fault activity.	WCET / resource stress measurement with worst-case traffic.
ZCA14-03	Invalid state, timing, resource or input information for POSIX Scheduling, WCET, CPU and Memory Reserve shall restrict the affected service or route and shall not permit stale configuration or uncontrolled transmit authority.	SIL/HIL timing, saturation, overload and invalid-input boundary sweep.

Verification, margin and future-change rule

Timing margin can disappear under synchronized network bursts, logging and fault handling even when nominal CPU utilization is low. Closure evidence shall include: WCET / resource stress measurement with worst-case traffic. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

FREEDOM FROM INTERFERENCE & SHARED RESOURCE ARBITRATION

Detailed subsystem PRD / engineering assurance annex - safety closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA15-01	The Freedom From Interference and Shared Resource Arbitration safety mechanism shall configuration-control hazard-relevant limits, independent detection and shutdown paths, FTTI assumptions, diagnostic coverage, safe/degraded states and external dependencies.	Architecture / ICD / configuration review.
ZCA15-02	Lower-criticality workloads shall not exceed their enforced resource budgets or access protected routing / safety state across the released shared-resource set.	Memory / DMA negative test plus CPU / interrupt / buffer exhaustion stress.
ZCA15-03	Loss of required margin, diagnostic validity or protection readiness for Freedom From Interference and Shared Resource Arbitration shall force the allocated safe/degraded routing state within the released FTTI without dependence on a lower-integrity workload.	Safety-mechanism fault injection with FTTI and independence evidence.

Verification, margin and future-change rule

FFI fails when any shared resource remains logically separated but physically unbounded or writable by a lower-criticality client. Closure evidence shall include: Memory / DMA negative test plus CPU / interrupt / buffer exhaustion stress. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

TIME BASE, EVENT ORDERING & STALE-TIMESTAMP HANDLING

Detailed subsystem PRD / engineering assurance annex - interface closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA16-01	The Time Base, Event Ordering and Stale-Timestamp Handling interface shall configuration-control signal ownership, units/range, update rate, freshness/integrity checks, timeout behavior, authority and non-permissive fallback.	Architecture / ICD / configuration review.
ZCA16-02	Loss, jump or invalidity of the controller time base shall not cause stale traffic to be accepted as fresh or fault logs to misorder safety-relevant events without detection.	Time jump / reset / stale-timestamp injection with routing and log review.
ZCA16-03	Invalid, stale or conflicting information for Time Base, Event Ordering and Stale-Timestamp Handling shall be treated as non-permissive; the controller shall restrict the affected route or service without granting additional network authority.	Protocol conformance with timeout, stale-data, range, integrity and authority fault injection.

Verification, margin and future-change rule

Routing, diagnostics and security can make inconsistent decisions if time validity is assumed rather than carried with the data. Closure evidence shall include: Time jump / reset / stale-timestamp injection with routing and log review. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

SECURE BOOT, HSM / ROOT OF TRUST & KEY STORAGE

Detailed subsystem PRD / engineering assurance annex - cybersecurity closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA17-01	The Secure Boot, HSM/Root of Trust and Key Storage security function shall configuration-control trust anchors, authenticated authority, key/configuration lifecycle, failure behavior, maintenance access and independence of local safety responses.	Architecture / ICD / configuration review.
ZCA17-02	Only authenticated, variant-compatible software / configuration shall reach normal routing-enabled state; invalid trust state shall remain non-permissive.	Modified-image, rollback, debug and key-store negative tests.
ZCA17-03	Loss or invalidity of trust for Secure Boot, HSM/Root of Trust and Key Storage shall reject unauthorized traffic or configuration changes while preserving immediate routing-integrity, watchdog and safe-state functions.	Negative-authorization / trust-loss test with concurrent safety-response monitoring.

Verification, margin and future-change rule

A centralized gateway concentrates authority, so compromise of boot trust can invalidate every higher-layer firewall and routing control. Closure evidence shall include: Modified-image, rollback, debug and key-store negative tests. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

MACsec SESSION, KEY ROTATION & LINK FAILURE

Detailed subsystem PRD / engineering assurance annex - cybersecurity closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA18-01	The MACsec Session, Key Rotation and Link Failure security function shall configuration-control trust anchors, authenticated authority, key/configuration lifecycle, failure behavior, maintenance access and independence of local safety responses.	Architecture / ICD / configuration review.
ZCA18-02	Protected Ethernet links shall reject unauthenticated / integrity-failed traffic and shall transition to the released degraded or non-permissive behavior on trust loss.	MACsec invalid-key, replay, rekey and link-reset test.
ZCA18-03	Loss or invalidity of trust for MACsec Session, Key Rotation and Link Failure shall reject unauthorized traffic or configuration changes while preserving immediate routing-integrity, watchdog and safe-state functions.	Negative-authorization / trust-loss test with concurrent safety-response monitoring.

Verification, margin and future-change rule

Link encryption is not sufficient if rekey, replay or trust-loss behavior allows stale or unauthorized traffic during transition. Closure evidence shall include: MACsec invalid-key, replay, rekey and link-reset test. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

HARDWARE FIREWALL, PORT POLICY & PRIVILEGE SEPARATION

Detailed subsystem PRD / engineering assurance annex - cybersecurity closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA19-01	The Hardware Firewall, Port Policy and Privilege Separation security function shall configuration-control trust anchors, authenticated authority, key/configuration lifecycle, failure behavior, maintenance access and independence of local safety responses.	Architecture / ICD / configuration review.
ZCA19-02	Hardware-firewall rules shall default to non-permissive behavior for protected traffic and shall not be modifiable by unauthorized lower-privilege software.	Policy corruption, privilege escalation and unauthorized-rule-write negative test.
ZCA19-03	Loss or invalidity of trust for Hardware Firewall, Port Policy and Privilege Separation shall reject unauthorized traffic or configuration changes while preserving immediate routing-integrity, watchdog and safe-state functions.	Negative-authorization / trust-loss test with concurrent safety-response monitoring.

Verification, margin and future-change rule

A firewall becomes a common point of failure if its management path is not more strongly controlled than the traffic it protects. Closure evidence shall include: Policy corruption, privilege escalation and unauthorized-rule-write negative test. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

OTA A/B UPDATE, ROLLBACK & POWER-FAIL INTEGRITY

Detailed subsystem PRD / engineering assurance annex - cybersecurity closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA20-01	The OTA A/B Update, Rollback and Power-Fail Integrity security function shall configuration-control trust anchors, authenticated authority, key/configuration lifecycle, failure behavior, maintenance access and independence of local safety responses.	Architecture / ICD / configuration review.
ZCA20-02	Interrupted update or failed activation shall preserve one authenticated bootable image and a compatible routing / firewall configuration set.	Power-interruption update / activation / rollback matrix.
ZCA20-03	Loss or invalidity of trust for OTA A/B Update, Rollback and Power-Fail Integrity shall reject unauthorized traffic or configuration changes while preserving immediate routing-integrity, watchdog and safe-state functions.	Negative-authorization / trust-loss test with concurrent safety-response monitoring.

Verification, margin and future-change rule

Image rollback alone is unsafe if routing policy, calibration or security metadata can remain at an incompatible revision. Closure evidence shall include: Power-interruption update / activation / rollback matrix. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

DIAGNOSTIC DATA, FIRST-FAULT CAPTURE & SERVICE ACCESS

Detailed subsystem PRD / engineering assurance annex - engineering closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA21-01	The Diagnostic Data, First-Fault Capture and Service Access design shall configuration-control the operating envelope, interfaces, ownership, limits, assumptions, fault response and objective release evidence for every released variant.	Architecture / ICD / configuration review.
ZCA21-02	Safety-relevant first-fault records shall preserve the released minimum context across controlled reboot without allowing diagnostic activity to bypass routing or FFI controls.	Fault injection, power-cycle retention and service authorization test.
ZCA21-03	If qualified margin or required health data for Diagnostic Data, First-Fault Capture and Service Access is unavailable, the controller shall inhibit or restrict the affected network/compute function and enter the allocated non-permissive or degraded behavior.	Boundary-condition and fault-injection / release-gate test with physical or network response recorded.

Verification, margin and future-change rule

Without first-fault preservation, repeated resets can erase the distinction between initiating fault and secondary network symptoms. Closure evidence shall include: Fault injection, power-cycle retention and service authorization test. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

PASSIVE HEATSINK THERMAL PATH & HOT-SOAK RESTART

Detailed subsystem PRD / engineering assurance annex - thermal closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA22-01	The Passive Heatsink Thermal Path and Hot-Soak Restart thermal design shall configuration-control heat-source limits, sensor locations, interface resistance, cooling/environment boundary, derating thresholds and end-of-life thermal margin.	Architecture / ICD / configuration review.
ZCA22-02	Hotspot temperatures shall remain below released limits for the full defined compute / network duty cycle including hot-soak restart and degraded thermal interface.	Thermal mapping and hot-soak restart with maximum traffic / CPU load.
ZCA22-03	If temperature validity or thermal margin for Passive Heatsink Thermal Path and Hot-Soak Restart is unavailable, the controller shall restrict the affected compute/network function and enter the allocated degraded state before device limits are exceeded.	Thermal / workload boundary test with derating or degraded-state response recorded.

Verification, margin and future-change rule

Passive cooling margin can vanish after hot soak or interface aging while network workload remains high. Closure evidence shall include: Thermal mapping and hot-soak restart with maximum traffic / CPU load. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

EMC / ESD / TRANSIENT IMMUNITY WITH ROUTING CONTINUITY

Detailed subsystem PRD / engineering assurance annex - hardware closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA23-01	The EMC/ESD/Transient Immunity With Routing Continuity design shall configuration-control its electrical operating envelope, component and interface limits, protection thresholds, tolerance assumptions and end-of-life margin for every released hardware variant.	Architecture / ICD / configuration review.
ZCA23-02	Qualified electrical / EMC disturbances shall not produce uncontrolled network transmission, route-table corruption or loss of required safety supervision.	Vehicle EMC / transient susceptibility test with full traffic and safety logging.
ZCA23-03	If qualified margin or required health data for EMC/ESD/Transient Immunity With Routing Continuity is unavailable, the controller shall inhibit or restrict the affected network/compute function and enter the allocated non-permissive or degraded behavior.	Representative hardware fault-injection test with network and safety response recorded.

Verification, margin and future-change rule

A network gateway may appear to recover functionally after an EMC event while having transmitted one hazardous transient frame burst. Closure evidence shall include: Vehicle EMC / transient susceptibility test with full traffic and safety logging. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

CONNECTOR, GROUNDING, HARNESS & HIGH-SPEED SIGNAL INTEGRITY

Detailed subsystem PRD / engineering assurance annex - hardware closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA24-01	The released connector, harness and PCB high-speed channel shall configuration-control topology, impedance and return-path assumptions, grounding and bonding, connector degradation, PI/SI limits, environmental variation and installation-specific acceptance margin.	Electrical and communication ICD plus SI/PI channel review.
ZCA24-02	The released connector / harness / PCB channel shall meet power integrity and high-speed link margin across tolerance, temperature, vibration and installation variation.	Channel simulation / measurement plus environmental correlation.
ZCA24-03	If SI/PI margin, harness or connector qualification, or environmental-correlation evidence is incomplete for a hardware or installation variant, that variant shall not be released until the required channel evidence is closed and approved.	SI/PI qualification closure review with channel measurement or simulation and environmental correlation.

Verification, margin and future-change rule

High-speed and power interfaces share a mechanical boundary; grounding or return-path changes can simultaneously degrade EMC and link stability. Closure evidence shall include: Channel simulation / measurement plus environmental correlation. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

ALUMINUM CHASSIS, ISO 16750-3 VIBRATION & STACK-UP

Detailed subsystem PRD / engineering assurance annex - mechanical closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA25-01	Mechanical qualification of the aluminum chassis and mounting stack-up shall configuration-control the ISO 16750-3 vibration profile, mounting interfaces, fastener and stack-up tolerances, thermal interfaces and post-stress acceptance limits for every released variant.	Mechanical interface and ISO 16750-3 vibration qualification-plan review.
ZCA25-02	Post-vibration and thermal-cycling functional checks shall confirm network, power, reset and safety-monitor performance within released limits.	ISO 16750-3 aligned vibration / mechanical test followed by electrical retest.
ZCA25-03	If vibration, thermal-cycle, mechanical stack-up or post-stress functional evidence is incomplete for a hardware or installation variant, that variant shall not be released until the required mechanical and environmental qualification is closed and approved.	Mechanical and environmental qualification closure review with post-stress electrical and network retest.

Verification, margin and future-change rule

Mechanical motion can create intermittent high-speed or power faults that are missed by static post-test inspection. Closure evidence shall include: ISO 16750-3 aligned vibration / mechanical test followed by electrical retest. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

BOUNDARY SCAN, ICT & PRODUCTION TEST ACCESS

Detailed subsystem PRD / engineering assurance annex - manufacturing closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA26-01	Production release for Boundary Scan, ICT and Production Test Access shall configuration-control critical characteristics, process capability, fixture/test coverage, measurement uncertainty, genealogy, calibration/provisioning and acceptance limits.	Architecture / ICD / configuration review.
ZCA26-02	Production structural tests shall cover the released accessible interconnect set and shall document alternate verification for safety-critical nets not observable by boundary scan / ICT.	Boundary-scan / ICT coverage report and golden-unit correlation.
ZCA26-03	If process capability, measurement traceability or required production evidence for Boundary Scan, ICT and Production Test Access is incomplete, production release of the affected variant shall remain blocked until the defined acceptance evidence is closed and approved.	Process-capability, gauge/fixture correlation, genealogy and approved production-release evidence.

Verification, margin and future-change rule

High coverage percentages can hide critical untested nets unless coverage is tied to product safety / routing significance. Closure evidence shall include: Boundary-scan / ICT coverage report and golden-unit correlation. A change to the process, qualification basis, supplier or acceptance method shall reopen the affected release gate before equivalence is claimed.

SERIALIZED MAC / IDENTITY PROVISIONING & CONFIGURATION

Detailed subsystem PRD / engineering assurance annex - manufacturing closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA27-01	Production release for Serialized MAC/Identity Provisioning and Configuration shall configuration-control critical characteristics, process capability, fixture/test coverage, measurement uncertainty, genealogy, calibration/provisioning and acceptance limits.	Architecture / ICD / configuration review.
ZCA27-02	Each production unit shall have unique serialized network identity and a traceable hardware / software / configuration manifest that prevents duplicate or incompatible release.	Provisioning audit, duplicate-ID negative test and genealogy check.
ZCA27-03	If process capability, measurement traceability or required production evidence for Serialized MAC/Identity Provisioning and Configuration is incomplete, production release of the affected variant shall remain blocked until the defined acceptance evidence is closed and approved.	Process-capability, gauge/fixture correlation, genealogy and approved production-release evidence.

Verification, margin and future-change rule

Duplicate identity or mismatched variant configuration can create field failures that are invisible to functional EOL testing. Closure evidence shall include: Provisioning audit, duplicate-ID negative test and genealogy check. A change to the process, qualification basis, supplier or acceptance method shall reopen the affected release gate before equivalence is claimed.

FULL-LOAD NETWORK ROUTING, FAILOVER & FAULT-INJECTION DVP

Detailed subsystem PRD / engineering assurance annex - verification closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA28-01	Verification release for Full-Load Network Routing, Failover and Fault-Injection DVP shall configuration-control representative boundary conditions, calibrated instrumentation, raw-data retention, configuration identity and objective pass/fail criteria.	Architecture / ICD / configuration review.
ZCA28-02	The DVP shall cover released network-load corners, credible link / compute / memory / protocol faults and relevant combined stresses with objective pass/fail criteria fixed before test.	Configuration-controlled network / HIL DVP with raw packet and timing records.
ZCA28-03	If the test matrix, instrumentation traceability, raw evidence or objective pass/fail disposition for Full-Load Network Routing, Failover and Fault-Injection DVP is incomplete, the verification gate shall remain open and no production-equivalence claim shall be made.	Configuration-controlled DVP report with calibrated instrumentation, raw records and explicit pass/fail disposition.

Verification, margin and future-change rule

Nominal bench traffic can miss queue, retry and timing interactions that appear only at combined load and fault corners. Closure evidence shall include: Configuration-controlled network / HIL DVP with raw packet and timing records. A change to the process, qualification basis, supplier or acceptance method shall reopen the affected release gate before equivalence is claimed.

ENVIRONMENTAL SEQUENTIAL STRESS & THERMAL / VIBRATION DVP

Detailed subsystem PRD / engineering assurance annex - verification closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA29-01	Verification release for Environmental Sequential Stress and Thermal/Vibration DVP shall configuration-control representative boundary conditions, calibrated instrumentation, raw-data retention, configuration identity and objective pass/fail criteria.	Architecture / ICD / configuration review.
ZCA29-02	Environmental qualification shall include powered or post-stress verification of routing, power, reset, safety monitor and high-speed interfaces for the released installation class.	Sequential environmental DVP with pre / during / post electrical comparison.
ZCA29-03	If the test matrix, instrumentation traceability, raw evidence or objective pass/fail disposition for Environmental Sequential Stress and Thermal/Vibration DVP is incomplete, the verification gate shall remain open and no production-equivalence claim shall be made.	Configuration-controlled DVP report with calibrated instrumentation, raw records and explicit pass/fail disposition.

Verification, margin and future-change rule

Independent environmental tests can miss cumulative solder, connector, thermal-interface and signal-integrity degradation. Closure evidence shall include: Sequential environmental DVP with pre / during / post electrical comparison. A change to the process, qualification basis, supplier or acceptance method shall reopen the affected release gate before equivalence is claimed.

15-YEAR OTA / NVM / OBSOLESCENCE & SECOND-SOURCE CLOSURE

Detailed subsystem PRD / engineering assurance annex - lifecycle closure for ASIL-D centralized zone controller.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ZCA30-01	Lifecycle closure for 15-Year OTA/NVM/Obsolescence and Second-Source Closure shall configuration-control mission profile, aging/endurance assumptions, field counters, supplier/obsolescence dependencies, evidence-reuse limits and regression triggers.	Architecture / ICD / configuration review.
ZCA30-02	Evidence reuse after component, software-platform, protocol or supplier change shall be permitted only inside documented similarity bounds; otherwise affected safety, cyber, timing, EMC and DVP evidence shall be reopened.	Lifecycle model, obsolescence plan and targeted change-impact regression review.
ZCA30-03	If lifecycle, reliability, similarity or change-impact evidence for 15-Year OTA/NVM/Obsolescence and Second-Source Closure is incomplete, the affected variant or second source shall not be approved until the required analysis and targeted regression are closed.	Lifecycle / reliability / similarity review with approved change-impact analysis and targeted regression evidence.

Verification, margin and future-change rule

Second-source and long-life changes can preserve nominal functionality while altering timing, thermal, EMC, security or diagnostic behavior that underpins earlier evidence. Closure evidence shall include: Lifecycle model, obsolescence plan and targeted change-impact regression review. A change to the process, qualification basis, supplier or acceptance method shall reopen the affected release gate before equivalence is claimed.

REV A2 RELEASE CLOSURE - ENGINEERING ASSURANCE BASELINE

Requirements, evidence, configuration and future-change closure.

The ZC-ASILD Rev A2 baseline combines the product-level architecture with a detailed system PRD and subsystem engineering-assurance annex. Release requires closure of vehicle-variant TBC values, reconciliation with the vehicle safety / cybersecurity lifecycle and objective evidence for each atomic requirement. No requirement is considered complete only because a related higher-level paragraph exists; completion is tied to the specified test, analysis, inspection, network trace, timing record or configuration evidence.

Gate	Evidence required / release blocker
Architecture / ICD	Released zone allocation, power variants, state / authority model, network topology and interfaces. Blocker: unbounded or conflicting ownership.
Safety / protection	Closed fault paths, FTTI / diagnostic coverage, FFI and ISO 26262 work-product traceability. Blocker: open high-severity mechanism or unverified independence.
HW / SW	WCCA, SI / PI, timing / WCET, memory protection, resource budgets and configuration integrity. Blocker: unverified derived requirement or margin deficit.
Cybersecurity	Secure boot, firewall / MACsec policy, trust / key lifecycle, OTA / rollback evidence. Blocker: unresolved high-risk threat path or unauthorized authority.
Verification	DVP with calibrated timing / network instrumentation, raw evidence and explicit pass/fail. Blocker: failed / waived critical test without approved rationale.
Manufacturing / lifecycle	Boundary-scan / EOL capability, serialized identity, mission profile, storage endurance and obsolescence controls. Blocker: process cannot demonstrate repeatable release margin or supported 15-year change equivalence.

Future-proof rule. Reuse of existing evidence after a processor, transceiver, routing-fabric, software-platform, vehicle-network or supplier change is allowed only inside documented similarity bounds. Any change affecting power integrity, timing / WCET, FFI, memory protection, high-speed signal integrity, cybersecurity authority, thermal resistance, reliability or interface ownership shall trigger targeted impact analysis and regression DVP before production equivalence is claimed.

Candidate standards / work products to confirm by program

- Baseline-defined: ASIL-D routing-integrity intent with strict Freedom From Interference; AUTOSAR Adaptive with POSIX-compliant OS; Ethernet backbone with hardware firewall and MACsec; ISO 16750-3 vibration; automated boundary scan and serialized MAC provisioning.
- Automotive safety / process candidates: ISO 26262 item / HARA / FSC / TSC and hardware / software safety work products, Automotive SPICE work products and program-specific OEM requirements.
- Network / platform candidates for confirmation: applicable IEEE 802.3 / 802.1AE MACsec, CAN FD / ISO 11898, LIN, PCI Express and AUTOSAR Adaptive platform / communication specifications for the released implementation.
- Vehicle electrical / environmental / EMC candidates: applicable ISO 16750, CISPR 25, ISO 11452, ESD / transient standards and OEM power / network / harness specifications.
- These candidate references require program confirmation; edition, scope and applicability must be confirmed by the responsible safety, cybersecurity, network and compliance owners.

RELEASE INTERPRETATION BOUNDARY

Any numeric value, vehicle-network allocation, silicon capability, supplier characteristic, timing budget or manufacturing threshold not explicitly released in this PRD remains a controlled program parameter. It shall be closed through the applicable ICD, parameter/configuration register, safety analysis, timing/WCCA, DVP or production release evidence before it is used as a compliance claim.